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Stock return forecast based on LSTM sentiment analysis

Abstract

This study aims to explore methods for stock return prediction and incorporates sentiment analysis to calculate sentiment values from comment data. Firstly, we compared the performance of dictionary-based methods, machine learning algorithms, and deep learning algorithms in sentiment analysis and selected the LSTM deep learning algorithm with higher accuracy for sentiment value calculation. Then, I combined the sentiment values with stock return prediction by aggregating and averaging the sentiment values and standardizing the relevant features to construct a feature set for stock price return prediction. I employed three methods, namely linear regression, random forest, and neural network, for return prediction and evaluated their performance. The experimental results demonstrate that sentiment analysis can provide valuable information for stock return prediction, and linear regression and random forest models perform well in terms of prediction performance. Specific evaluation metrics include mean squared error (MSE), root mean squared error (RMSE), and mean absolute error (MAE). The findings of this study provide decision-making references for investors and offer feasible approaches for decision-making in the financial domain.

Keywords: stock return prediction; sentiment analysis; machine learning algorithms; deep learning algorithms; LSTM; neural network

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1. Research Background and Significance

1.1. Research Background

(1)Public Opinion Information Plays a Significant Role in Modeling

In the modern financial market, successful investors are adept at utilizing high-quality information for investment decisions. More importantly, they are capable of making prompt and effective decisions based on the information they have acquired. Bondt, a financial scholar, has proposed that stock price fluctuations are influenced by irrational investor psychology and behavior, which encompass macroeconomic conditions, current policies, corporate strategies, and investor sentiment. The first three factors have long-term effects on the entire industry and experience minimal significant changes within short periods, making it challenging to predict the direction of the stock market based on them. However, when facing complex and uncertain stock market investment decisions, investor sentiment is susceptible to internal and external influences, such as public news, rumors, short-term market fluctuations, and other investors' opinions. Furthermore, emotional changes can be dramatic. With the proliferation and development of the internet, an increasing number of people obtain information through online sources. The 51st Statistical Report on the Development of China's Internet issued by the China Internet Network Information Center (CNNIC) reveals that by December 2022, the number of internet users in China had grown to 1.067 billion, with an internet penetration rate of 75.6%. The impact of online public opinion on investor decisions is becoming increasingly significant. However, public opinion information is voluminous, posing challenges for manual processing. Additionally, this information is time-sensitive, necessitating swift decision-making in the financial market. Therefore, it is imperative to achieve automation in information processing, enabling investment decisions to be based on automatically generated recommendations that comprehensively consider all public opinion messages relevant

to a particular financial asset.

(2) New Algorithms Demonstrate Advantages in Stock Price Prediction

In recent years, the advancement of technologies such as Natural Language Processing (NLP) and neural networks has provided new research avenues for utilizing stock price data and commentary data to predict stock returns. With the widespread use of the internet and social media, a large amount of commentary data, including news reports, expert opinions, investor sentiments, and discussions on social media, has become available. These commentary data provide insights into the perspectives and sentiment tendencies of market participants, exerting significant influence on stock market trends. NLP techniques assist researchers in extracting useful information from these textual data, such as sentiment analysis, topic modeling, and keyword extraction, to better understand the emotions and opinions of market participants.

Simultaneously, neural networks, as powerful machine learning methods, have the capacity to learn and extract patterns from large-scale data and possess strong nonlinear modeling capabilities. Through training neural networks, models adaptable to the stock market can be constructed, with stock price data and commentary data used as inputs to predict stock price trends. The deep structure and complexity of neural networks allow for better capturing of nonlinear relationships and temporal features in the market, thereby improving prediction accuracy.

Furthermore, with the development of big data technologies, I are better equipped to process and analyze large-scale stock price and commentary data. The combination of NLP techniques and the capability to handle big data enables fast processing and analysis of massive textual data, uncovering information and patterns within. This provides more robust data support and computational power for utilizing stock price

data and commentary data in stock price prediction.

1.2. Significance of the Research

Performing sentiment analysis on textual data and integrating it with stock price data for stock return prediction offers a comprehensive, accurate, and effective support for stock price forecasting and trading decisions. It holds significant research significance and practical application value. The specific significance is as follows:

(1) Considering sentiment factors: Sentiment analysis helps capture market participants' sentiment tendencies towards specific stocks or related news events. By conducting sentiment analysis on social media, news reports, forum posts, and other textual data, one can obtain sentiment values of market participants towards a particular stock, enabling a more comprehensive understanding of the impact of market sentiment and investor emotions on stock prices.

(2)Providing additional features: Combining sentiment analysis results with stock price data can provide additional features for stock return prediction. The results of sentiment analysis can be used as one of the input features for stock price prediction models, combined with traditional technical indicators, fundamental data, etc., to provide more comprehensive and multi-dimensional information, thereby enhancing the accuracy and robustness of prediction models.

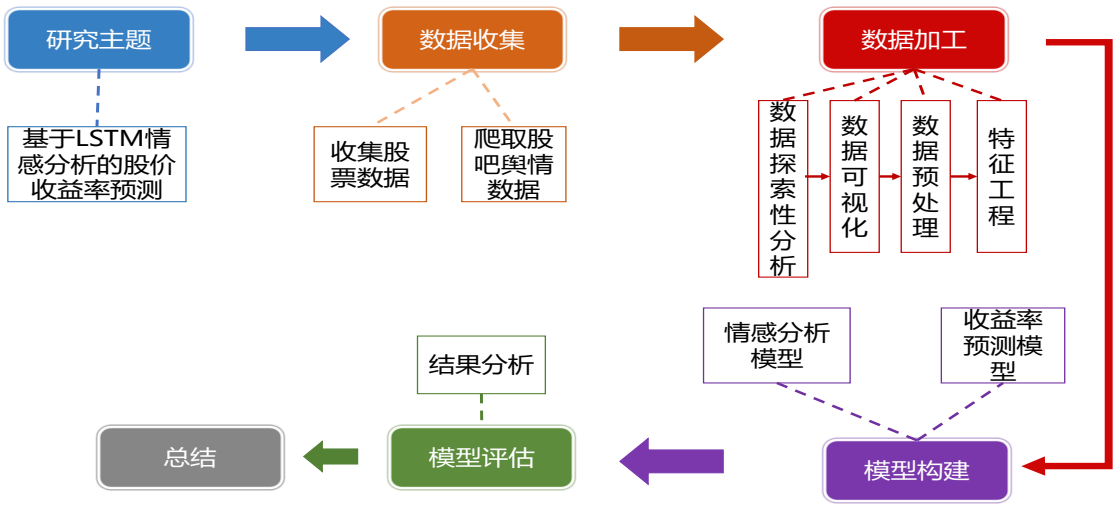
(3) Capturing market sentiment changes: Sentiment analysis helps identify changes and turning points in market sentiment. Investor sentiment is often one of the driving factors behind market fluctuations, and sentiment analysis can assist in monitoring and capturing changes in market sentiment, providing timely market signals and decision-making references for investors.

(4) Enhancing the effectiveness of trading strategies: Integrating sentiment analysis with stock return prediction can be applied to the design and optimization of quantitative trading strategies. By considering sentiment factors, more intelligent and adaptable trading strategies can be developed, improving the portfolio's return and risk

management capabilities.

2. Research Framework

The research framework of this study is as follows



3. Data Description and Preprocessing

3.1.Data Sources

The stock price data of the Shanghai Stock Exchange Composite Index (SHCI) from January 1, 2021, to May 15, 2023, was obtained through the public API qstock and saved as a CSV file. The stock price data typically includes information such as dates, opening prices, highest prices, lowest prices, closing prices, and trading volumes. These data can be used to calculate stock returns, technical indicators, and more. Analyzing historical stock price data enables an understanding of market trends, volatility, and market participants' trading behavior, providing a basis for investment decisions. By acquiring and storing the stock price data of the SHCI, it can be combined with other data, such as sentiment analysis results, for predicting stock

returns in future research.

With the popularity of the internet and mobile communication devices, an increasing number of individuals choose to express their opinions and emotions on social media platforms, resulting in a substantial amount of commentary and public opinion texts. These texts contain people's views, predictions, and emotions regarding the stock market, which are of significant importance for stock market analysis and forecasting. In this research, the Oriental Fortune Network platform was selected as the target platform. Python web scraping and data extraction tools, such as Octoparse（八爪鱼采集器）, were used to collect the stock market commentary data of the SHCI from January 1, 2021, to May 15, 2023. In total, 382,464 data points were collected during this period. These data will be used to analyze market sentiment and trends, as well as predict future stock prices.

3.2.Data introduction

(1) Stock Data

Stock data includes the stock price and trading volume data for each stock on each trading day, comprising eight dimensional attributes. Among them, one is a temporal attribute, and five are numerical attributes. The symbols and explanations of these attribute fields are as follows:

Table 1 Stock data attribute names and descriptions

Attribute name	attribute specification
Date	Trading date
Open	opening price
High	highest price
Low	lowest price
Close	closing price
Volume	volume of business

(2)Comment Data

Through the web crawler, a total of 382,464 comment data related to the CSI 300 index contains 3 attributes for each piece of data, namely: title, date, number of likes, number of replies, number of retweets and website. The data format of a single piece of data as follows:

Table 2 comments on data property names and descriptions

link	https://guba.eastmoney.com/o/news,zssh000001,1109206656.html
Title	券商暴动大金融带动所有板块集体爆发啦，牛市启动啦，才卖
Time	2021-11-19 13:45:11
Author	交易之大道至简
Likes	3
Comments amount	0
content	满仓干啊

(3) Comment data with emotional labels

Data with sentiment labels were used for model training for sentiment analysis. This part of the data contains a total of 7118, comments are from the stock, artificial emotional label classification. Each comment contains two properties, comment, sentiment_label. The data format of a single piece of data as follows:

Table 3 comments on data property names and descriptions

Comment	这个培圾，天天有几万手卖盘。这个培圾，天天有几万手卖盘。。。
sentiment_label	-1

4. Data Exploration and Variable Selection

4.1.descriptive statistics

Table 4 Descriptive statistical table

	链接	标题	作者	日期	点赞	评论	内容
count	273116	273116	273116	273116	273116	273116	273116
unique	273116	268489	71526	865	26.6685	30.9060	268673
top	https://guba.eastmoney.com.	大幅上调上证指数目标点位至 8565.05点[胜利]	威廉欧奈尔	2022/4/25	0	0	
freq	1	52	1895	1328	2295	9280	244

As can be seen from the descriptive table, the comments captured included a total of 71,526 users who post some contents, and the user who posted the most comments posted 1,895 comments. The most frequent author is "William O 'Neil" and the most frequent date is "2022-04-25".

"Like", "comment" two attributes unique, top, freq are respectively the variance, minimum to maximum. The number of likes was 2,295, and the number of comments

was 9,280. The mean value of the comment column is 2.602729 and the standard deviation is 30.906006. Similar to the likes column, the average item receives about 2 comments, but the standard deviation is large, indicating a large variation in the number of comments.

Table 5 Descriptive statistics of stock price data

	open	high	low	close	volume	turnover	turnover_rate
count	571	571	571	571	5.71E+02	5.71E+02	571
mean	3364.43	3385.0	3343.33	3366.3	3.39E+08	4.28E+11	0.785377
std	190.628	188.975	190.335	190.306	7.18E+07	9.17E+10	0.166394
min	2866.82	2926.02	2863.65	2886.43	1.89E+08	2.27E+11	0.44
25%	3229.95	3253.13	3212.07	3233.51	2.92E+08	3.65E+11	0.68
max	3721.09	3731.69	3692.82	3715.37	6.67E+08	8.43E+11	1.55

From the above descriptive statistics, it can be found that during the period from 2021 to May 16, 2023, the closing price of the Shanghai Composite Index is at least 2886.43 and at most 3692.82, with great fluctuations. The average of the opening price, high price, low price, and closing price is not much different, indicating that the

price fluctuation in a single day is relatively small.

4.2.data preprocessing

(1) Data integration: Since the comment data is partly obtained by the python crawler and partly by the octopus collector, it is necessary to merge the data for subsequent processing. At the same time, the data attribute is renamed.

(2) outlier processing: Due to the structure of the web page, some data will appear to be incomplete links, attributes corresponding to anomalies, etc., so it is necessary to delete these abnormal data. A total of 382,426 pieces of data were captured and 362,571 pieces of data were processed.

(3) Processing of missing values: Some data comments were empty. Due to sufficient data, the missing values were deleted, leaving 273,116 pieces of data.

Data conversion: The captured date data contains the specific time when the

(4) comments were published, so the data format such as "2021-11-19 13:45:11" needs to be converted to "2021-11-19", and then the comments are reordered

according to the date.

4.3. Data visualization and feature engineering

(1)Visualization of stock price data

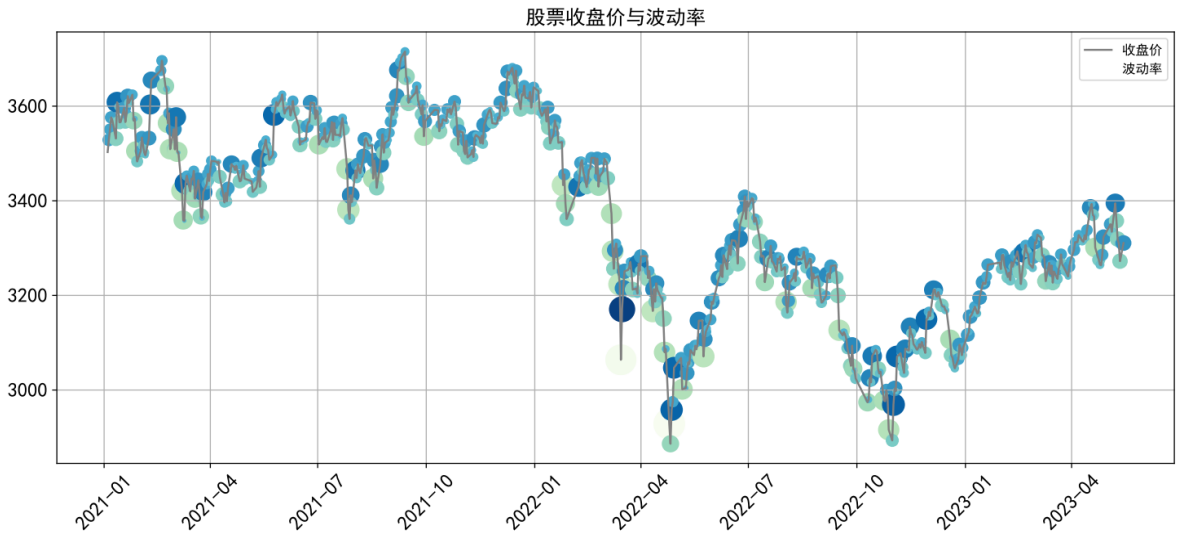


Figure 1 Visualization of closing prices and volatility

The above chart shows the closing price change of the Shanghai Composite Index from 2022-01 to 2023-05 through a discount chart, and shows the size transformation of the yield in the form of a scatter chart. The size and color of the scatter point are determined by the size of the return rate, so that the change of the stock price return rate can be directly observed. It can be seen that from January 2021 to April 2022, the closing price of the Shanghai Composite Index showed a downward trend, a slight increase from April 2022 to July 2022, a downward trend from July 2022 to October 2022, and then an upward trend.

(2)Information visualization of the author of the published content

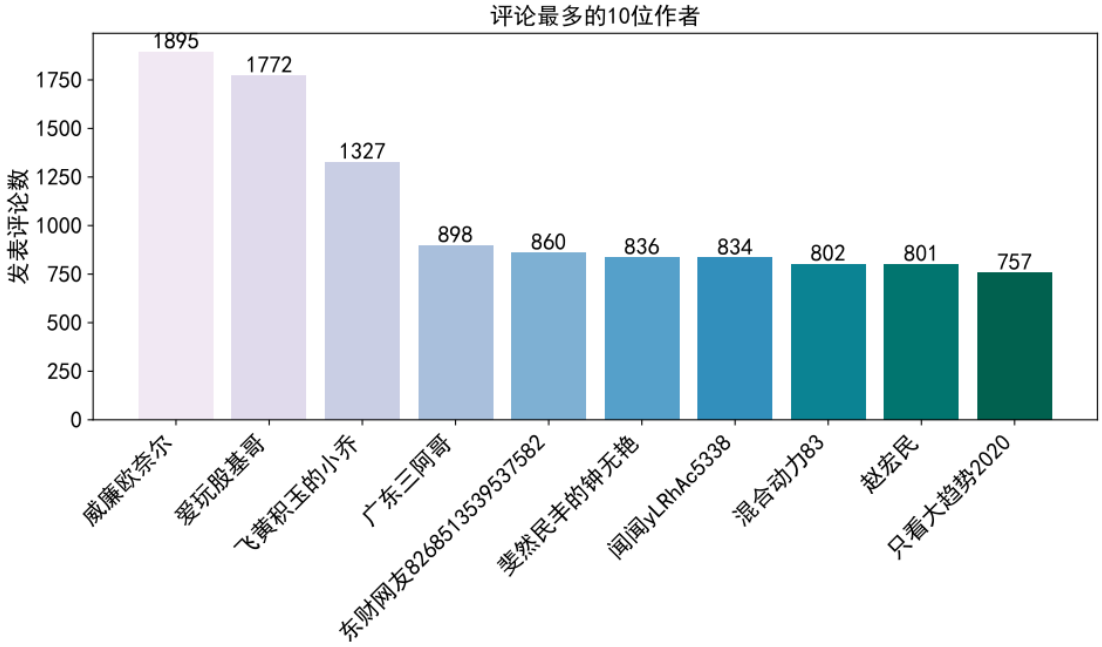


Figure 2 The authors statistical bar chart

As can be seen from the bar chart above, among the top 10 authors who posted the most comments, the user named "William O 'Neil" posted the most, with 1,895 comments. This shows that the author of the comment is concerned about the stock price of the Shanghai Composite Index. Among them, the number of posts posted by authors who published less was 757, indicating that the number of posts was relatively high.

(3) The visualization of the statistical values

In order to use the number of comments and likes of each post and the total number of daily posts for stock price return forecast, the average number of comments, the average number of likes and the number of daily comments were counted, and the average number of comments and the average number of likes of related features were constructed as characteristics.

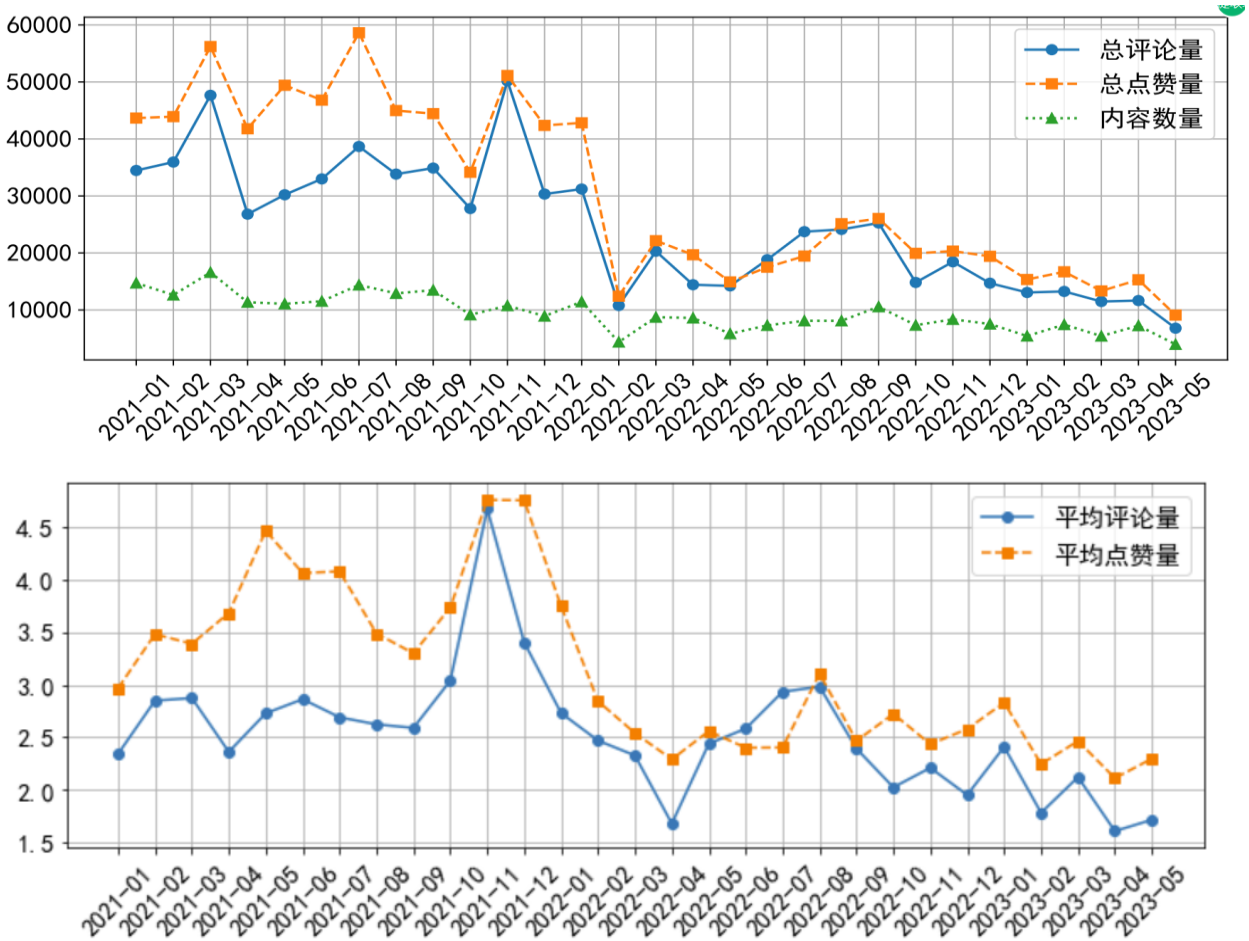


Figure 3 Comment and like the discounted graph

From the above discount chart, it can be seen that the total number of comments and the total number of likes as well as the average number of comments and the average number of likes are highly similar, and the fluctuations of these four characteristics are relatively similar to the fluctuations of stock prices. The number of posts fluctuated less than the other two variables, but it also showed a similar trend to the stock price fluctuation. Therefore, the average number of comments, the average number of likes and the number of posts obtained by statistics can be used as the characteristics of stock price return prediction.

The following 3 D scatter chart can further reflect the average number of comments and evaluate the impact of likes on the stock price.

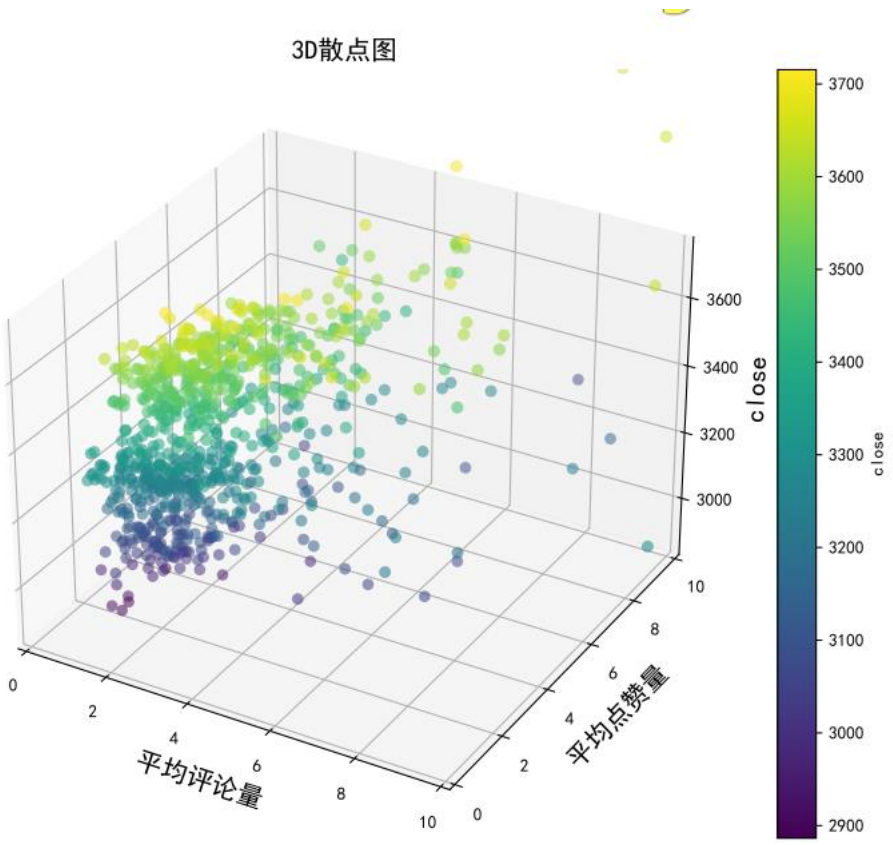


Figure 4 3D graph of comment and like

It can be seen that the more the average number of comments and likes, the higher the Shanghai Composite Index will close.

Methodology

5.1.Emotion analysis model

Common methods for sentiment analysis include dictionary-based analysis, machine learning algorithms, and deep learning algorithms. In order to choose the method with higher accuracy for analysis, three kinds of methods are tried respectively. The data for model training is the review data that has been manually annotated, a total of 7118 pieces.

Table 6 Comparison of effect of emotion analysis model

Method	Accuracy	F1 Score	Recall	Precision
based_dict	0. 768334	0. 658945	0.663474	0.654478
MultinomialNB	0.848315	0.733990	0.656388	0.832402
SGDClassifier	0.856039	0.772979	0. 730126	0.821176
LSTM	0.879213	0. 806306	0. 788546	0.824885

(1)Dictionary-based method (based_dict)

The financial dictionary compiled by Teacher Li Qing's team was used, which contained 1523 positive dictionaries and 2025 negative dictionaries respectively. Dictionary-based sentiment analysis is a common approach to determine the emotional polarity of a comment by matching the words in the comment to the sentiment dictionary. Based on the given results, the dictionary-based approach performs 0.768, 0.659, 0.663, and 0.654 for accuracy, F1 Score, recall, and accuracy, respectively. Although the dictionary-based approach is less accurate, given its simplicity and ease of implementation, it can still be used as a method for rapid preliminary analysis. Some of the emotional labels run are as follows:

```
午后光线会给大家一个惊喜吗午后光线会给大家一个惊喜吗
{'sentences': 1, 'words': 16, 'pos': 12, 'neg': 0}
1
逃命要紧逃命要紧
{'sentences': 1, 'words': 2, 'pos': 0, 'neg': 0}
{'sentences': 1, 'words': 2, 'pos': 0, 'neg': 0}
-1
今天大盘就要回调了，小光线到顶了么？[加油]今天大盘就要回调了，小光线到顶了么？
{'sentences': 2, 'words': 27, 'pos': 13, 'neg': 16}
-1
浪费时间，在这个股上浪费时间，在这个股上
{'sentences': 1, 'words': 10, 'pos': 0, 'neg': 15}
-1
继续买买，把最后子弹射出去继续买买，把最后子弹射出去
{'sentences': 1, 'words': 14, 'pos': 0, 'neg': 11}
-1
```

(2)Machine learning algorithm

Sentiment analysis classification is performed using MultinomialNB and SGDClassifier. Sentiment analysis is performed using two machine learning algorithms MultinomialNB and SGDClassifier. According to the results, the accuracy is 0.848 MultinomialNB algorithm, F1 Score is 0.734, recall rate is 0.656, and accuracy is 0.832. The SGDClassifier algorithm performed slightly better in terms of accuracy, reaching 0.856, F1 Score 0.773, recall rate 0.730, and accuracy rate 0.821. Compared with dictionary-based methods, these machine learning algorithms can better capture the complexity and context of the review data, improving the accuracy of sentiment analysis.

(3)Deep Learning algorithm

LSTM is a commonly used deep learning algorithm for processing sequential data, such as text. According to the results, the LSTM algorithm achieved 0.879 accuracy, 0.806 F1 Score, 0.789 recall rate, and 0.825 accuracy rate. Compared with machine learning algorithm, LSTM algorithm has stronger modeling ability and expression ability, and can better capture semantic and contextual information in text, thus improving the accuracy of sentiment analysis. Part of the training is as follows:

```
Epoch 1/10
178/178 [==============================] - 136s 755ms/step - loss: 0.5114 - accuracy: 0.7457
Epoch 2/10
178/178 [==============================] - 134s 754ms/step - loss: 0.2484 - accuracy: 0.9071
Epoch 3/10
178/178 [==============================] - 135s 756ms/step - loss: 0.3039 - accuracy: 0.8895
Epoch 4/10
178/178 [==============================] - 86s 483ms/step - loss: 0.1842 - accuracy: 0.9378
Epoch 5/10
178/178 [==============================] - 93s 522ms/step - loss: 0.1281 - accuracy: 0.9631
Epoch 6/10
178/178 [==============================] - 90s 509ms/step - loss: 0.0966 - accuracy: 0.9737
Epoch 7/10
178/178 [==============================] - 83s 466ms/step - loss: 0.0760 - accuracy: 0.9793
Epoch 8/10
178/178 [==============================] - 82s 458ms/step - loss: 0.0624 - accuracy: 0.9824
Epoch 9/10
178/178 [==============================] - 82s 463ms/step - loss: 0.0524 - accuracy: 0.9851
Epoch 10/10
178/178 [==============================] - 83s 467ms/step - loss: 0.0452 - accuracy: 0.9861
```

5.2.Rise and fall prediction model

According to the results of the sentiment analysis model, the LSTM model with higher accuracy is adopted to conduct sentiment analysis on the sample data, and the emotion values of different posts are obtained. In order to apply the sentiment analysis of the text to the stock return prediction, the sentiment values are summed according to the date, and the average value of the sentiment values is obtained in order to eliminate the effect of the number of posts. Then, in order to eliminate the impact of dimension, the closing price, the daily average number of comments, the daily average number of likes, and the daily number of posts are standardized, which are used as characteristics for stock price return prediction. In order to predict the specific rate of return, linear regression, random forest and neural network models are applied to continue the prediction.

Three prediction methods (linear regression, random forest and neural network) were evaluated and their performance and performance were analyzed. Common indicators including Mean Squared Error (MSE), Root Mean Squared Error (Root Mean Squared Error) were calculated. RMSE and Mean Absolute Error (MAE) were analyzed.

Table 6 Comparison of effect of Rise and fall prediction model

Method	MSE	RMSE	MAE
linear regression	0.00007	0.00813	0.00625
random forest	0.00007	0.00848	0.00651
neural network	0.00014	0.01172	0.0083

Based on the results of the above evaluation indicators, the following conclusions and analyses can be drawn:

(1) On a given data set, linear regression and random forest models perform similarly and better than neural network models. Linear regression and random forest models

have low MSE, RMSE and MAE values, which indicates that they can fit the data and make predictions more accurately.

(2) The performance of neural network model is slightly inferior to that of linear regression and random forest model. The MSE, RMSE, and MAE values of the neural network model are all higher, which may mean that it does not fit the data as well as the other two models.

(3) When evaluating the model, in addition to considering numerical indicators, the complexity and computational cost of the model should also be considered. Linear regression models have simple forms and lower computational costs, random forest models combine multiple decision trees and have higher accuracy and certain computational costs, while neural network models require larger computational resources and more training time.

5. Conclusion

The aim of this study is to explore methods for stock return prediction and incorporate sentiment analysis for calculating sentiment values from comment data. Firstly, we employed three common sentiment analysis methods: dictionary-based method, machine learning algorithms, and deep learning algorithms. By comparing their accuracy and performance, I selected the LSTM deep learning algorithm with higher accuracy for sentiment analysis. This method demonstrated good performance in sentiment value calculation on the sample data.

In applying sentiment analysis to stock return prediction, I aggregated the sentiment values by date and calculated the average value to mitigate the impact of post quantity. Additionally, to eliminate the dimensionality effect, I standardized the closing price, daily average number of comments, daily average number of likes, and daily number of posts, which were used as features for stock price return prediction.

I utilized three methods: linear regression, random forest, and neural network, to predict stock returns and evaluated their performance using metrics such as mean

squared error (MSE), root mean squared error (RMSE), and mean absolute error (MAE). Based on the evaluation results, linear regression and random forest models exhibited similar performance and outperformed the neural network model. This suggests that linear regression and random forest models can more accurately fit the data and make predictions, while the neural network model had relatively poorer fit.

In conclusion, this study effectively predicted stock returns by combining sentiment analysis and feature engineering with different prediction models. I found that sentiment analysis provides value for stock return prediction, and linear regression and random forest models perform well in terms of prediction performance. The findings not only provide insights for investors but also offer a feasible approach for decision-making in the financial domain. However, further research is needed to optimize and improve the prediction models for enhanced accuracy and stability.